FIKR Model — Lesson Plan

Learning Leading to Production

Title	Completing the Square	**Subject / Unit**	Topic 2 — Quadratic Functions and Equations · Lesson 2-5
Grade	Grade 10 — Algebra II	**Date**	Week 7 · Semester 1, 2026–27
Format	[ ] 40-minute [X] 60-minute	**Teacher**	Mr Malek Thiab

Essential Question (curriculum map)

Not shown — the curriculum map's Essential Question column for Topic 2 was not in the project text, so it is left open rather than invented. Add it when confirmed.

Lesson Objectives (curriculum map — verbatim)

1. Rewrite a quadratic equation into the form $(x - p)^2 = q$ by completing the square.
2. Determine the minimum or maximum value of a quadratic function using completing the square.
3. Solve real-world problems modeled by quadratic equations using completing the square.

Vocabulary (curriculum map) · Assessments (curriculum map)

Vocabulary: adjustment term ; completing the square.

Assessments: “Give it a go” questions during practice ; “Time to Check” at the Mastery Gate.

0 Preparation Before Class

Learning Outcome: What will the student be able to do?

Students will find the adjustment term and rewrite $x^2 + bx$ as a perfect square (objective 1; HSA.REI.B.4.A).

Students will solve quadratic equations by completing the square, including $a \neq 1$ and complex solutions (objective 1; HSA.REI.B.4.B and HSN.CN.C.7).

Students will find the minimum or maximum value of a quadratic function and where it occurs (objective 2; HSA.SSE.B.3.B and HSF.IF.C.8.A).

Students will solve a real-world problem modeled by a quadratic equation (objective 3; HSF.LE.B.6).

The curriculum map’s Essential Question and Math Practices columns for Topic 2 were not in the project text; they are left open, not invented.

Three teaching days this week. If time runs short, cut Smart Production and carry it into the next session — never the Mastery Gate, which is the evidence.

Pre-Class Preparation — Topic/Link and Evidence of Readiness: What will the student complete before class?

Flipped video (4 min): “Making a perfect square” — why $(x + 5)^2$ has the middle term $10x$. Posted on the LMS

48 hours ahead.

Two embedded EdPuzzle questions: expand $(x+5)^2$, and factor x^2+6x+9.

Evidence of readiness: EdPuzzle completion report plus both embedded answers submitted before the bell.

1 Diagnostic 6 min

Tool Used: What evidence indicates differences in students' current levels?

3-question Quizizz (≤ 4 minutes) — the slide-5 checks: expand $(x+4)^2$; factor $x^2+10x+25$; solve $(x-3)^2=16$.

Answers: $x^2+8x+16$; $(x+5)^2$; $x=7$ or $x=-1$. No calculator.

Where is the learning gap? What will you adjust in your instruction accordingly?

Expected gap 1 — Q1: the middle term is forgotten and $(x+4)^2$ becomes x^2+16; completing the square is this expansion run backwards.

Expected gap 2 — Q3: students give only $x=7$ and lose the minus root.

Routing: 0–1 correct → re-teach expanding a squared bracket and square roots with plus and minus. 2 → straight to the adjustment term. 3 → move quickly through block 1 and begin the Investigate prompt early.

2 Targeted Instruction 18 min

Required Instruction and Modeling: What needs direct explanation? Which example will demonstrate the thinking process and the expected quality standard?

Slide 6 — Perfect squares and the adjustment term (objective 1; HSA.REI.B.4.A). $x^2+bx+\left(\frac{b}{2}\right)^2=\left(x+\frac{b}{2}\right)^2$.

Examples: $x^2+10x+25$; $x^2-7x+\frac{49}{4}$.

Slide 7 — Solve by completing the square (objective 1; HSA.REI.B.4.B). Add the adjustment term to BOTH sides.

Examples: $x^2+6x-7=0$ gives $x=1$ or -7; $x^2-8x+5=0$ gives $x=4\pm\sqrt{11}$; $x^2+4x+13=0$ gives $x=-2\pm3i$.

Slide 8 — When the coefficient of x^2 is not 1 (objective 1). Divide every term by a first. Examples:

$2x^2-12x+7=0$ gives $x=3\pm\frac{\sqrt{22}}{2}$; $3x^2+12x+15=0$ gives $x=-2\pm i$.

Slide 9 — Maximum and minimum values (objective 2; HSA.SSE.B.3.B). $y=a(x-h)^2+k$. Examples: $x^2-6x+11$ has minimum 2; $-2x^2+8x+1$ has maximum 9; $3x^2+12x+7$ has minimum -5.

Narration focus — whatever is added to one side of an equation must be added to the other.

Quick Check for Understanding: What question or evidence confirms that students are ready to move on to practice?

On whiteboards, 60 seconds: complete the square for $x^2+6x+\square$ and for $x^2-8x+\square$.

Expected: 9 and 16. Watch for 36 and 64 (squaring b instead of half of b) and for 6 and 8.
80% correct → release guided practice. Below 80% → one more modelled example with a different coefficient.

3 Practice 12 min

Guided Task: How will the student progress from guided to independent practice? What feedback will the student use?

Two "Give it a go" problems, identical for every student, worked together with the teacher:
(1) Complete the square: $x^2 + 12x + \square$. (2) Solve $x^2 - 6x - 16 = 0$ by completing the square.
Answers: (1) 36, giving $(x+6)^2$. (2) $(x-3)^2 = 25$, so $x = 8$ or $x = -2$.
Feedback: worked live on the board step by step; students self-mark and circle their OWN first error.

Independent Task and Feedback Tool: students choose their route and may switch at any time

Delivered through the "Choose Your Route" sheet with live feedback in Pear Deck. Routes are named for the task, not for the student.

PRACTICE — Secure the method. Complete three squares, solve $x^2 + 12x + 20 = 0$ and $x^2 - 10x + 24 = 0$, and find the minimum of $x^2 - 4x + 9$. Done when: the adjustment term is added to both sides and every solution is checked by substitution.

APPLY — Use it in context. Solve $2x^2 + 8x - 5 = 0$, find the maximum height of a model water jet, and the greatest area of a rectangle with a fixed perimeter. Done when: the leading coefficient is divided out first, and the extreme value and where it occurs are both stated with units.

INVESTIGATE — Find out why. Why the adjustment term is added to both sides; how the quadratic formula comes from $ax^2 + bx + c = 0$; why the vertex has $x = -\frac{b}{2a}$. Done when: your argument keeps the equation balanced at every step.

Every student sits the same Mastery Gate. The gate is not differentiated, because it is the evidence.

4 Production — Stage One 12 min

Production Task: What new task, situation, or product will allow the student to demonstrate learning?

Non-routine transfer task — this context has not appeared in practice (slide 11).

A model water jet on the Jeddah Corniche, a Red Sea waterfront project that supports Vision 2030 tourism: $h(t) = -5t^2 + 30t + 2$ metres, t seconds after the jet leaves the nozzle. The model values are invented for practice.

(a) Complete the square and state the greatest height and when it happens. (b) Find the two times at which the height is 42 m. (c) Find the time at which the jet reaches sea level, to the nearest hundredth of a second.

Answers: (a) $h = -5(t-3)^2 + 47$: 47 m at $t = 3$ s. (b) $(t-3)^2 = 1$, so $t = 2$ s or $t = 4$ s. (c) $(t-3)^2 = \frac{47}{5}$, so $t = 3 + \sqrt{9.4} \approx 6.07$ s (the negative root is rejected).

Students then hand their reasoning to the AI critic.

Product Quality Criteria and the Role of AI: What is the minimum acceptable standard? What must the student complete independently?

Minimum acceptable standard: the factor -5 taken out and the adjustment term balanced, the maximum and its time both stated, the negative root rejected with a reason, and units (metres, seconds) in every answer.

Role of AI: critic only. Briefed prompt — "Challenge my reasoning and point out any step I have not justified." Never "solve this".

Completed independently by the student: the written justification for parts (a)–(c). AI may challenge it; it may not write it.

5 Presentation — Evidence of Learning 6 min

Evidence of Learning and Success Criterion: What will the student present or explain, and how? What must be clearly demonstrated for learning to be considered achieved?

"Time to Check" — one individual task. No notes, no partner, no AI.

1. Complete the square: $x^2 - 10x + \square$. 2. Solve $x^2 + 8x - 9 = 0$ by completing the square. 3. Find the minimum value of $f(x) = x^2 + 6x + 4$.

Answers: 1. 25, giving $(x-5)^2$. 2. $(x+4)^2 = 25$, so $x = 1$ or $x = -9$. 3. $(x+3)^2 - 5$, so the minimum is -5.

Done when: the adjustment term is added to both sides in question 2 and question 3 states the value, not the x-coordinate.

Scored live in Pear Deck as submissions arrive; result logged in the Impact Dashboard Mastery Rate.

6 Final Product — Refinement 6 min

Refinement and Final Product: What revision will the student make based on feedback? What will the final product look like?

Students revise their water-jet reasoning using the AI critique and the teacher's micro-conference note.

Final product, three equally acceptable formats: a Canva one-pager, a photographed hand-written solution, or a 45-second voice note.

Required content: the vertex form, the maximum and its time, the two times at 42 m; ONE worked non-example — writing $x^2 + 6x = 7$ as $(x+3)^2 = 7$ by adding 9 to the left side only — with a sentence on why it must be added to both sides; and a caption on why the jet has two times at the same height.

Graded on originality and clarity of reasoning, not correctness alone.

Presentation/Publication and Reflection: Where will the product be presented or published? What reflection question will close the learning cycle?

Published to the class LMS portfolio; the strongest go on the Mathematics Department wall as the Topic 2 modelling set.

Reflection question: "Which is harder for you — finding the adjustment term, or keeping the equation balanced?"

Total minutes across the seven stages: 60 (6 + 18 + 12 + 12 + 6 + 6) — matches the 60-minute format, including transitions.

★ Required Pillars

Differentiation · Adaptive Learning · Saudi Connection · Exam Alignment

Differentiation — three routes named for the task; students choose, and may switch at any time

PRACTICE — Secure the method. Complete the square, solve and find a minimum. Done when: the adjustment term is added to both sides.

APPLY — Use it in context. $a \neq 1$, a water jet and a rectangle of greatest area. Done when: the extreme value and where it occurs are both stated.

INVESTIGATE — Find out why. Why add to both sides, the quadratic formula and the vertex. Done when: the argument keeps the equation balanced.

No student is labelled. The routes describe tasks; every student sits the same Mastery Gate.

Adaptive Learning — how the lesson branches on evidence

Entry branch — the Quizizz result routes each student to the re-teach table, straight to the new content, or the Investigate prompt.

Mid-lesson branch — Pear Deck flags error patterns during independent practice; flagged students get a 2-minute micro-conference while others continue.

Exit branch — the Mastery Gate routes each student: PASS → Enrichment & Challenge; NOT YET → Targeted Learning Clinic at the start of the next lesson.

Routing is communicated privately through the LMS, never read aloud.

Saudi Culture & Real-World Connection

Production and project context: a model water jet on the Jeddah Corniche on the Red Sea (invented model values), a Vision 2030 tourism setting.

Discussion prompt: why do designers need the greatest height of a jet and the time it takes to come back to the sea?

SAT / SAAT (Tahsili) / GAT (Qudurat) Connection

SAT — Advanced Math: the minimum value of $y = x^2 - 6x + 5$ is -4. Trap: answering 3 (the x-coordinate) or -9 (forgetting to add 5).

GAT (Qudurat) — which is greater, the minimum of $x^2 - 4x + 7$ or 2? The minimum is 3, so the first. Trap: picking the x-value 2.

SAAT (Tahsili) — the solutions of $x^2 + 6x + 2 = 0$ are $-3 \pm \sqrt{7}$. Trap: adding 3 instead of 9 to both sides.

Suggestions for Improvement: